NIFTY 50 Dataset
(Timestamp, OHLC Data)

Data Preprocessing :-
- Convert timestamp
- Remove closed hours
- Sort data
- Compute log return

Volatility Calculation :-
- Rolling Std Dev
- EW Volatility
- ATR
- Intraday Range %

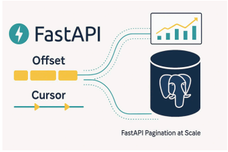
Spike Detection Logic :-
- Mean + 2σ rule
- Percentile threshold

Regime Classification :-
- Low
- Normal
- High

Store in PostgreSQL

PostgreSQL

FastAPI Backend :-
- /volatility
- /spikes
- /summary
- /regime-classification



React Dashboards :-
- Charts
- Spike Highlights
- Filters
- Regime visuals